MODEL QUESTION PAPER

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

THIRD SEMESTER B. TECH DEGREE EXAMINATION, MONTH AND YEAR

Course Code: GAEST305

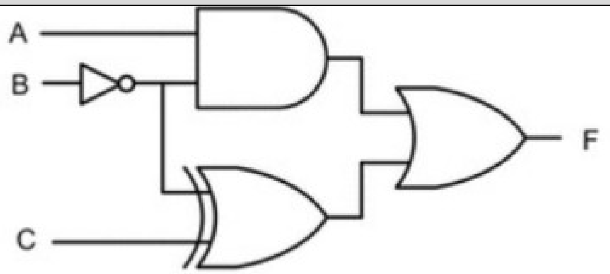
Course Name: DIGITAL ELECTRONICS AND LOGIC DESIGN

Max. Marks: 60 | Duration: 2 hours 30 minutes

PART A

		Answer all questions. Each question carries 3 marks	CO	Marks
1		Realize the Boolean expression F=A’BC+A’B’+BC using NOR gates alone.		(3)
2		A weather monitoring station records the temperature as 8-bit signed integer using twos complement representation. If the current temperature is -5 degrees, obtain its two's complement representation. If the temperature increases by 12 degrees, obtain the new temperature after performing the binary addition using two's complement method.		(3)
3		You are given a task to design an industrial press machine’s safety system. The press will operate (output F=1) only under the following safe conditions. i. The master switch M should be on with the emergency switch either on or off. ii. Two protective guards A and B must be either simultaneously on or either one of them should be on. Create the truth table describing the output F, based on the inputs M,E,A,B. Deduce the simplified Boolean expression in SOP form for the security system		(3)
4		Design a 3 bit code converter circuit with a mode control bit ‘M’, which converts the input binary code to gray code when M=0, and a gray code to		(3)

		its equivalent binary code when M=1. Draw the logic diagram.		
5		Design and realize an 8:1 multiplexer using two 4:1 multiplexers with necessary truth tables and explanations.		(3)
6		Realize a full subtractor circuit using a 3:8 decoder.		(3)
7		Derive the characteristic table, characteristic equation, and excitation table for a JK flipflop		(3)
8		Design an asynchronous mod 5 up counter using JK flipflop. Draw its timing diagram and explain.		(3)
		PART B		
		Answer any one full question from each module. Each question carries 9 marks		
		Module 1		
9	a)	Obtain the following (i) Hexadecimal equivalent of the octal number 745 (ii) Decimal equivalent of the hexadecimal number F52.2B (ii) Binary equivalent of the decimal number 75.925		(5)
	b)	A logic circuit family has all its gates (NAND, INVERTER, NOR)working with the following DC specifications: Output High Voltage (VOH) = 4.5V Output Low Voltage (VOL) = 0.5V Input High Voltage (VIH) = 3.0V Input Low Voltage (VIL) = 1.0V Power Supply (VDD) = 5V Determine the Noise Margin Low and Noise Margin High for this circuit family.		(4)
1	a)	Compute $64_{10} - 100_{10}$ using 8-bit two's complement addition. Provide the 8-bit result and indicate whether two's complement overflow occurred.		(4)

0		Check your result by converting the 8-bit result back to decimal.		
	b)	You are using an 8-bit signed fixed-point number system with a **Q3.4** format (3 integer bits, 4 fractional bits, 1 sign bit). a. What is the largest positive decimal number that can be represented in this Q3.4 format? b. What is the smallest (most negative) decimal number that can be represented in this Q3.4 format? c. Represent the decimal number `5.75` in this 8-bit Q3.4 format. Show your steps.		(5)
Module 2				
1 1	a)	 (i) For the given combinational circuit, determine the logic expression for F. (ii). Obtain the truthtable for F. (iii) Deduce and realize the circuit using NOR gates only. (iv). Deduce and realize the circuit using NAND gates only.		(6)
	b)	Simplify the following Boolean expressions to a minimum number of literals and draw the logic diagrams of the circuits that implement the simplified expressions (a). $a'bc + abc' + abc + a'bc'$ (b). $wxy'z + w'xz + wxyz$		(5)

12	a)	Using K-map reduction obtain the minimized sum of products (SOP) expression for F=m(1,3,9,11)+d(0,4,5). Also obtain the POS expression for F		(6)
	b)	Write the verilog module for realizing the SOP and POS expressions for F in above question using using the conditional operator (?) in dataflow model.		(3)
Module 3				
13	a)	Design and realize a full adder circuit using two half adders.		()
	b)	Write the complete verilog module for a 2:1 multiplexer using the conditional operator (?) in dataflow model. The module should have two data input 'in0', 'in1', and one selection input 'sel' and one output 'out'.		()
	c)			()
14	a)	Design a logic circuit to realize the boolean expression F=\sum m(0, 1, 2, 5, 6, 8, 10) using (i). an 8:1 multiplexer, assuming that the multiplexer works with logic low enable input and produces logic low outputs (ii) 1:16 demultiplexer, assuming that the demultiplexer works with logic low enable input and produces logic low outputs.		()
	b)	Design a simple digital circuit that implements a 4 bit unsigned adder with an overflow indicator. The design should adhere to the modern digital design flow using Verilog HDL emphasizing good abstraction.		()
Module 4				
15	a)	Design a synchronous sequence generator to generate the sequence 0,2,4,6,0,2,4,6,....		(4)
	b)	Write the behavioural Verilog code to implement a negative edge triggered D flipflop and hence develop the behavioural model to implement a 4 bit synchronous up counter by instantiating the D flipflop.		(5)

1 6	a)	Design a mod 10 synchronous counter using JK flipflop. Draw the the circuit and its timing diagram.		(6)
	b)	Design and develop the logic diagram to convert a D flipflop into a JK flipflop.		(3)
*****				